

Mondrian Conformal Prediction for Uncertainty Quantification in ER Discrete-Event Simulation Surrogates

Full Project Results (through Week 15)

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Problem Statement

- Surrogate models are fast, learned approximations of expensive simulations, increasingly used to guide decisions in stochastic service systems - ER staffing, queueing networks, etc.
- A point prediction alone isn't enough for a decision like "will adding 2 more doctors bring wait time down enough?" - that needs a calibrated interval, not just a number.
- Gaussian Processes are the established way to get one, but are expensive to train and rely on distributional assumptions rather than a finite-sample guarantee.
- Conformal Prediction (CP) is distribution-free with a finite-sample coverage guarantee and is cheap to calibrate - but it's only been validated for surrogate models in a narrow set of domains so far.

Research Gap

Gopakumar et al. (2026) validate Conformal Prediction for surrogate-model UQ —
but only in physics domains: PDEs, magnetohydrodynamics (MHD), weather, fusion.

The paper states two explicit limitations:

- Marginal coverage only — no guarantee of coverage within specific subgroups/conditions.
- Relies on an exchangeability assumption between calibration and test data — untested under distribution shift.

Neither limitation has been tested outside physics simulation. Discrete-event / queueing systems are a fundamentally different domain — stochastic discrete arrivals and departures, resource contention, priority scheduling — not continuous PDE fields. This is the gap this project sits in.

Why Mondrian CP

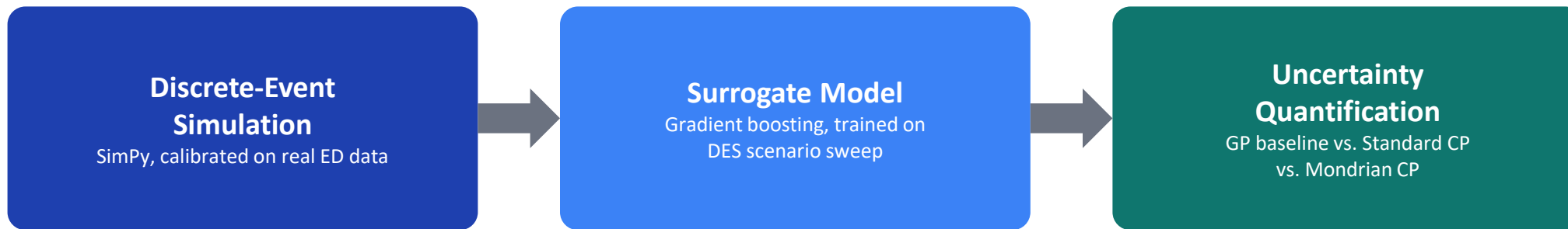
The problem with standard CP

- Standard CP calibrates one pooled quantile across every scenario in the calibration set — only guarantees coverage on average.
- A pooled interval is calibrated on average difficulty, so it can badly fail exactly the scenarios that matter most (understaffed + high demand), while wasting width on easy ones (overstaffed + low demand).
- Averaged coverage can look fine while specific operating conditions quietly fail.

What Mondrian CP does about it

- Calibrates a separate quantile per category — here, staffing tercile x arrival-rate tercile — instead of one pooled quantile.
- Each category gets its own, individually honest coverage guarantee instead of borrowing strength from categories that don't resemble it.
- Directly targets Gopakumar et al.'s marginal-coverage limitation — the first place it's been tested outside physics simulation.

Methodology Overview



Each stage validated against the previous before moving on:

- DES output validated against real aggregated ED statistics (91.0% match to real daily volume).
- Surrogate accuracy validated against DES outputs held out from training (MAE / RMSE / R^2).
- GP baseline, standard CP, Mondrian CP all evaluated on the same fixed test split and target coverage ($\alpha = 0.1$), so every method is directly comparable.
- Standard/Mondrian CP calibrated on a separate 1200-scenario set — never the surrogate's own training data, since reusing training residuals would understate the true residual spread.

Real-World Data Used

Hospital Triage and Patient History Data (Kaggle, maalona)

Yale New Haven Health System, retrospective study, March 2014 – July 2017

560,486

total ED visits

972

variables per visit

3

EDs (1 academic + 2 community)

258.2 / day

calibrated rate, Dept. A

What we extracted — real vs. literature-calibrated (disclosed explicitly, not blended silently):

From real data

- Arrival pattern by 4-hour bin, day of week, month
- ESI acuity mix (triage severity distribution)
- Department-level daily visit rate

From literature (not in dataset)

- Service/treatment time per ESI level (log-normal)
- Dataset has no discharge timestamp — confirmed by
 - exhaustive search across all 972 columns

DES Validation

Metric (200 simulated days)	Value
Real visits/day (Dept. A)	258.2
Simulated mean visits/day	235.1
Match	91.0%

The ~9% shortfall is expected, not a calibration flaw: patients still queued when a simulated 24h day ends are right-censored out of that day's stats rather than carried into a "next day," since each run is meant to be one independent sample for surrogate training, not a continuous rollout. This censoring effect resurfaces later in the exchangeability stress test (Slide 13).

Surrogate Model Accuracy

Target	MAE	RMSE	R ²
n_patients	9.90	12.59	0.929
mean_wait_minutes	8.86	13.23	0.787
mean_total_minutes	9.88	13.61	0.762
p95_wait_minutes	66.94	102.47	0.647

Gradient boosting (HistGradientBoostingRegressor), trained on 5000 DES scenarios sweeping staffing and arrival-rate combinations.
p95_wait_minutes fits worst — a tail statistic from one noisy simulated day depends heavily on the specific stochastic realization, not just the two scenario parameters. This matters later in the exchangeability stress test.

GP Baseline (the benchmark)

Target	Target coverage	Empirical coverage	Mean width
n_patients	90%	88.5%	39.3
mean_wait_minutes	90%	87.7%	43.6
mean_total_minutes	90%	88.8%	43.4
p95_wait_minutes	90%	90.1%	350.3

GP undercovers on 3 of 4 targets — expected, since it relies on distributional assumptions rather than a finite-sample guarantee. This is the gap the rest of the project tests whether CP / Mondrian CP can close.

Standard Conformal Prediction

Target	Sym. coverage	Sym. width	Asym. coverage	Asym. width
n_patients	92.1%	43.6	92.0%	43.6
mean_wait_minutes	89.0%	47.2	88.7%	47.2
mean_total_minutes	90.2%	46.4	89.6%	47.2
p95_wait_minutes	90.8%	362.9	89.7%	355.6

Two nonconformity measures tested: symmetric $|y - \hat{y}|$, and an asymmetric version using separate upper/lower residual quantiles for skewed targets. CP already lands closer to the 90% target than GP, with no systematic undercoverage bias — but it's still a single pooled quantile, so it only guarantees coverage on average, not within any specific scenario.

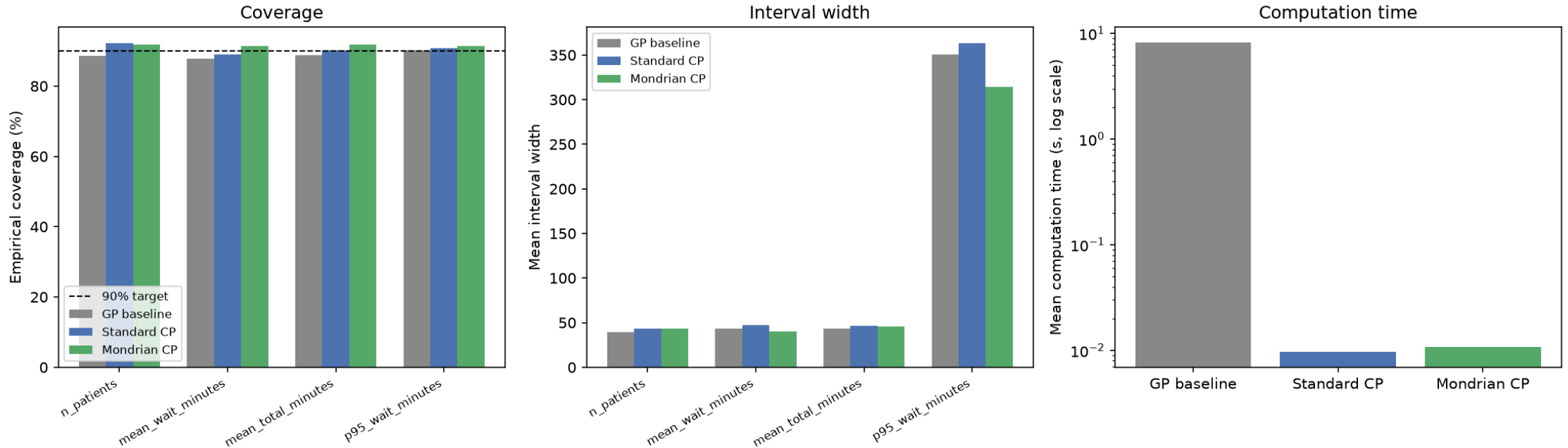
Mondrian CP Closes the Gap

Target	Pooled coverage (staff=Low / arrival=High)	Mondrian coverage (same category)	Target
mean_wait_minutes	68.2%	90.9%	90%
mean_total_minutes	80.7%	92.0%	90%
p95_wait_minutes	72.7%	92.0%	90%

The worst pooled-CP category is consistently staff=Low/arrival=High — understaffed during a demand surge, the scenario that matters most operationally. A pooled quantile is calibrated on average difficulty, so it silently fails exactly this scenario while overcovering the easy one (staff=High/arrival=Low hits 100% pooled coverage — wasted width). Mondrian gives the hard category an honestly wider interval instead of pretending it's as easy as the rest.

n_patients is the one target where Mondrian does not help — its pooled coverage was already fairly uniform across categories (87–99%), so there was no real conditional gap to fix, and splitting into 9 smaller calibration cells just adds finite-sample noise. That's an honest, expected Mondrian CP tradeoff, not a failure.

GP vs. Standard CP vs. Mondrian CP



CP calibrates ~650–1000x faster than GP (~0.01s vs. 7–11.5s). Mondrian CP's marginal coverage matches or beats standard CP's despite ~9x smaller per-category calibration slices, and is also usually narrower — pooling was wasting width in easy categories.

Exchangeability Stress Test

Arrival multiplier	n_patients	mean_wait	mean_total	p95_wait
1.3 (in range)	93.3%	83.0%	88.0%	82.0%
1.5	78.7%	68.7%	73.7%	69.7%
1.8	70.3%	42.7%	47.3%	43.0%
2.0	64.7%	34.3%	39.0%	33.0%
3.0	31.7%	12.3%	4.7%	78.3%*

Standard CP shown (Mondrian tracks closely). Root cause, verified directly: the gradient-boosting surrogate cannot extrapolate past its training range — mean_wait_minutes's prediction is frozen at 42.4 for every multiplier from 1.3 to 3.0 while the true value keeps climbing.

*p95_wait's apparent recovery at 3.0x isn't the interval working — it's the Slide 7 day-boundary censoring effect getting worse at extreme overload (fewer patients finish within a day, shrinking the completed-visit pool the p95 is computed over).

Both stated Gopakumar et al. limitations are now tested in this domain.

What We Accomplished

- Built and validated an ER discrete-event simulation (SimPy) calibrated on real hospital arrival data — 91.0% match to real daily patient volume.
- Generated a 5000-scenario training sweep and trained a surrogate model with strong accuracy on 3 of 4 targets (R^2 0.76–0.93).
- Implemented a GP baseline, standard conformal prediction (two nonconformity measures), and Mondrian conformal prediction, all evaluated on the same test split and target coverage.
- Demonstrated Mondrian CP closing a real marginal-vs-conditional coverage gap (68–81% pooled coverage in the hardest category, corrected to 91–92%), and quantified where it doesn't help.
- Ran an exchangeability stress test across 8 severity levels and root-caused the coverage collapse to a specific, verified mechanism, not just "it broke."
- Full head-to-head comparison of all three UQ methods on coverage, interval width, and computation time.

Did We Meet Our Novelty Target?

The claim was to test Gopakumar et al. (2026)'s two stated CP limitations — marginal coverage, exchangeability — outside the physics domains they validated CP in, using a discrete-event / queueing simulation surrogate instead.

- Marginal coverage limitation: tested. Mondrian CP closes the gap where a real gap exists (3 of 4 targets), and the one target where it doesn't help is explained, not hidden.
- Exchangeability limitation: tested. Confirmed it breaks down outside the training distribution, with a specific, verified root cause rather than a generic "performance degrades" statement.

Yes — both limitations were tested in a domain they hadn't been tested in before, with concrete, quantified answers for each, not just a replication of the base paper's physics-domain results.

Limitations & What's Left

- Service/treatment time is literature-calibrated, not derived from the dataset — the dataset has no discharge timestamp. Disclosed explicitly throughout, not blended silently into the "real data" story.
- Calibrated on one department (the largest/academic site) of one health system — generalization to other hospitals or EDs is untested.
- The exchangeability breakdown is partly a surrogate-model artifact (tree-based extrapolation), not purely a CP phenomenon — a different surrogate architecture might behave differently OOD.
- Literature review (30 papers, 3 in depth) still in progress.
- This is a working results deck, not the final end-sem PPT — that will follow the instructor's specific guidelines once shared.